

Q1 - 24 January - Shift 2

A Spherical ball of radius 1mm and density 10.5 g/cc is dropped in glycerine of coefficient of viscosity 9.8 poise and density 1.5 g/cc. Viscous force on the ball when it attains constant velocity is 3696×10^{-x} N. The value of x is

(Given, $g = 9.8 \text{ m/s}^2$ and $\pi = \frac{22}{7}$)

Space for your notes:

Q2 - 25 January - Shift 2

A spherical drop of liquid splits into 1000 identical spherical drops. If u_i is the surface energy of the original drop and u_f is the total surface energy of the resulting drops, the (ignoring evaporation).

$\frac{u_f}{u_i} = \left(\frac{10}{x}\right)$. Then value of x is _____ :

Space for your notes:

Q3 - 29 January - Shift 1

Surface tension of a soap bubble is $2.0 \times 10^{-2} \text{ Nm}^{-1}$.

Work done to increase the radius of soap bubble from 3.5 cm to 7 cm will be : [Take $\pi = \frac{22}{7}$]

(1) $0.72 \times 10^{-4} \text{ J}$

(2) $5.76 \times 10^{-4} \text{ J}$

(3) $18.48 \times 10^{-4} \text{ J}$

(4) $9.24 \times 10^{-4} \text{ J}$

Space for your notes:

Q4 - 29 January - Shift 2

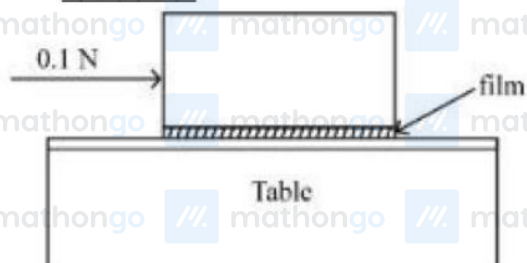
A fully loaded boeing aircraft has a mass of 5.4×10^5 kg. Its total wing area is 500 m^2 . It is in level flight with a speed of 1080 km/h . If the density of air ρ is 1.2 kg m^{-3} , the fractional increase in the speed of the air on the upper surface of the wing relative to the lower surface in percentage will be ($g = 10 \text{ m/s}^2$)

- (1) 16 (2) 6
(3) 8 (4) 10

Space for your notes:

Q5 - 29 January - Shift 2

A metal block of base area 0.20 m^2 is placed on a table, as shown in figure. A liquid film of thickness 0.25 mm is inserted between the block and the table. The block is pushed by a horizontal force of 0.1 N and moves with a constant speed. If the viscosity of the liquid is $5.0 \times 10^{-3} \text{ Pl}$, the speed of block is $\text{---} \times 10^{-3} \text{ m/s}$.



Space for your notes:

Q6 - 30 January - Shift 1

The height of liquid column raised in a capillary tube of certain radius when dipped in liquid A vertically is, 5 cm . If the tube is dipped in a similar manner in another liquid B of surface tension and density double the values of liquid A, the height of liquid column raised in liquid B would be $\text{---} \text{ m}$.

- (1) 0.20 (2) 0.5
(3) 0.05 (4) 0.10

Space for your notes:

Q7 - 31 January - Shift 1

If 1000 droplets of water of surface tension 0.07 N/m, having same radius 1 mm each, combine to form a single drop. In the process the released surface energy is-

(Take $\pi = \frac{22}{7}$)

(1) 7.92×10^{-6} J

(2) 7.92×10^{-4} J

(3) 9.68×10^{-4} J

(4) 8.8×10^{-5} J

Space for your notes:

Q8 - 01 February - Shift 1

A mercury drop of radius 10^{-3} m is broken into 125 equal size droplets. Surface tension of mercury is 0.45 Nm^{-1} . The gain in surface energy is:

(1) 2.26×10^{-5} J

(2) 28×10^{-5} J

(3) 17.5×10^{-5} J

(4) 5×10^{-5} J

Space for your notes:

Q9 - 01 February - Shift 2

The surface of water in a water tank of cross section area 750 cm^2 on the top of a house is h m.

above the tap level. The speed of water coming out through the tap of cross section area 500 mm^2 is

30 cm/s. At that instant, $\frac{dh}{dt}$ is $x \times 10^{-3}$ m/s. The value of x will be .

Space for your notes:

Answer Key

Q9 (2)

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Q1 (7)

When the ball attain terminal velocity

$$F_v = (mg - F_B) \quad (\because a = 0)$$

$$= V\sigma_b g - V\rho_\ell g$$

$$= Vg(\sigma_b - \rho_\ell)$$

$$= \frac{4}{3}\pi(10^{-3})^3 \times 9.8 (10.5 - 1.5) \times 10^3$$

$$= 3696 \times 10^{-7} \text{ N}$$

$$\text{So, } x = 7$$

Q2 (1)

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Surface Tension = T

R : Radius of bigger drop

r : Radius of smaller drop

Volume will remain same

$$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3$$

$$R = 10r$$

$$u_i = T \cdot 4\pi R^2$$

$$u_f = T \cdot 4\pi r^2 \times 1000$$

$$\frac{u_f}{u_i} = \frac{1000r^2}{R^2}$$

$$\frac{u_f}{u_i} = \frac{10}{1}$$

$$\frac{u_f}{u_i} = \frac{10}{1}$$

$$\text{So, } x = 1$$

Q3 (3)

Surface area of soap bubble = $2 \times 4\pi R^2$

Work done = change in surface energy $\times T_s$

$$= T_s \times 8\pi (R_2^2 - R_1^2)$$

$$= 2 \times 10^{-2} \times 8 \times \frac{22}{7} \times 49 \times \frac{3}{4} \times 10^{-4}$$

$$= 18.48 \times 10^{-4} \text{ J}$$

Q4 (4)

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$$P_2 A - P_1 A = 5.4 \times 10^5 \times g$$

$$P_2 - P_1 = \frac{5.4 \times 10^6}{500} = 5.4 \times 2 \times 10^2 \times 10$$

$$= 10.8 \times 10^3$$

$$P_2 + 0 + \frac{1}{2} \rho V_2^2 = P_1 + 0 + \frac{1}{2} \rho V_1^2$$

$$P_2 - P_1 = \frac{1}{2} \rho (V_1^2 - V_2^2) = \frac{1}{2} \rho (V_1 - V_2)(V_1 + V_2)$$

$$10.8 \times 10^3 = \frac{1}{2} \times 1.2 (V_1 - V_2) \times 2 \times 3 \times 10^2$$

$$10.8 \times 10 = 3.6 (V_1 - V_2)$$

$$V_1 - V_2 = 30$$

$$\left(\frac{V_1 - V_2}{V} \right) \times 100 = \frac{30}{300} \times 100 = 10\%$$

Q5 (25)

$$|F| = \eta A \frac{\Delta v}{\Delta h} : 0.1 = 5 \times 10^{-3} \times 0.2 \times \frac{v}{.25 \times 10^{-3}}$$

$$v = 0.025 \text{ m/s or } v = 25 \times 10^{-3} \text{ m/s}$$

Q6 (3)

$$h = \frac{2S \cos \theta}{r \rho g}$$

$$\therefore \frac{h_1}{h_2} = \frac{S_1 \rho_2}{S_2 \rho_1}$$

$$\frac{5}{h_2} = \left[\frac{1}{2} \right] \left[\frac{2}{1} \right] \Rightarrow h_2 = 5 \text{ cm} = 0.05 \text{ m}$$

{Info about angle of contact not there so most

Q7 (2)

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$$1000 \times \frac{4\pi}{3} (1)^3 = \frac{4\pi}{3} R^3$$

$$R = 10 \text{ mm}$$

$$T \times 1000 \times 4\pi (10^{-3})^2 - T \times 4\pi (10 \times 10^{-3})^2 = \Delta E$$

$$\Delta E = 4 \times \pi \times 7 \times 10^{-2} [1000 - 100] \times 10^{-6}$$

$$\Delta E = 7.92 \times 10^{-4} \text{ J}$$

Option 2.

Q8 (1)

$$\text{Initial surface energy} = 0.45 \times 4\pi (10^{-3})^2$$

$$\frac{4}{3} \pi (10^{-3})^3 = 125 \times \frac{4\pi}{3} R_{\text{new}}^3$$

$$\therefore 10^{-3} = 5 R_{\text{new}}$$

$$\therefore R_{\text{new}} = \frac{10^{-3}}{5} \text{ m}$$

$$\text{So, final surface energy} = 0.45 \times 125 \times 4\pi \left(\frac{10^{-3}}{5} \right)^2$$

$$\text{Increase in energy} = 0.45 \times 4\pi \times (10^{-3})^2 \left[\frac{125}{25} - 1 \right]$$

$$= 4 \times 0.45 \times 4\pi \times 10^{-6}$$

$$= 2.26 \times 10^{-5} \text{ J}$$

Q9 (2)

$$A_1 V_1 = A_2 V_2$$

$$750 \times 10^{-4} V_1 = 500 \times 10^{-6} \times 0.3$$

$$V_1 = \frac{500 \times 3 \times 10^{-3}}{750} \text{ m/s}$$

$$= 2 \times 10^{-3} \text{ m/s}$$

$$\frac{dh}{dt} = -2 \times 10^{-3} \text{ m/s}$$

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